

# Text and Image as Data

## AI and Economics Summer School

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- Last block: predictive models are fit, regularized, tuned, and evaluated out of sample
- **First:** how raw text and images become vectors, and what the geometry of those vectors means
- **Part I: Text.** Representation, the three families of method, topic models, and what goes wrong when documents come from different environments
- **Part II: Images.** Classification, CNNs, what replaced them, and joint text/image spaces
- Two running research applications, both with the same structure: *measurement first, then economics*

# Outline

- 1 Representations and Embeddings
- 2 Text as Data
- 3 Image as Data
- 4 Application: Partisanship and Attention in Visual Media
- 5 Wrap-up

# An Embedding Is a Learned Map

Start from a raw object  $r_i$ —a sentence, document, image, or video—that lives in some complicated space  $\mathcal{X}$ . An encoder maps it into a fixed-length vector:

$$f_{\theta} : \mathcal{X} \rightarrow \mathbb{R}^d, \quad z_i = f_{\theta}(r_i) \in \mathbb{R}^d.$$

- $f_{\theta}$  is a neural network;  $\theta$  is learned on a large pretraining corpus
- $d$  is fixed—typically hundreds or thousands—even when raw objects have different lengths or sizes
- The vector  $z_i$  is the **embedding**. Its coordinates are not usually interpretable one by one
- Once we have  $z_i$ , the methods from the last block apply directly:

$$\hat{y}_i = g_{\beta}(z_i), \quad \text{or} \quad \{z_i\}_{i=1}^N \xrightarrow{\text{PCA / clustering}} \text{lower-dimensional structure.}$$

# The Information Is in the Geometry

Normalize each embedding to unit length:

$$\tilde{z}_i = \frac{z_i}{\|z_i\|}.$$

Then cosine similarity is just a dot product:

$$\text{sim}(i, j) = \frac{z_i^\top z_j}{\|z_i\| \|z_j\|} = \tilde{z}_i^\top \tilde{z}_j, \quad d_{\cos}(i, j) = 1 - \text{sim}(i, j).$$

- Similar objects should lie close together; unrelated objects should lie far apart
- A group can be summarized by its centroid,  $\bar{z}_G = |G|^{-1} \sum_{i \in G} z_i$
- A direction can become a measure: project  $z_i$  onto a concept vector  $v$  using  $z_i^\top v$
- Absolute cosine values have no natural interpretation. Comparisons, rankings, and differences do

## How the Geometry Is Learned

Suppose  $(a_i, b_i)$  is a matched pair: a word and its context, two versions of the same sentence, or an image and its caption. Two encoders produce normalized vectors

$$u_i = \frac{f_\theta(a_i)}{\|f_\theta(a_i)\|}, \quad v_j = \frac{g_\phi(b_j)}{\|g_\phi(b_j)\|}.$$

A contrastive loss makes the true pair more similar than the mismatched pairs in the batch:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(u_i^\top v_i / \tau)}{\sum_{j=1}^N \exp(u_i^\top v_j / \tau)}.$$

- The numerator pulls matched pairs together; the denominator pushes mismatched pairs apart
- Word2Vec uses word–context pairs. CLIP uses image–caption pairs. The learning problem is the same

# From Tokens to a Document Embedding

A transformer returns one contextual vector per token:

$$H_i = F_\theta(w_{i1}, \dots, w_{iT_i}) = (h_{i1}, \dots, h_{iT_i}), \quad h_{it} \in \mathbb{R}^d.$$

The token vector depends on the entire sequence, so the embedding of “bank” changes with its context. To represent the whole document, pool the token vectors:

$$z_i = \text{pool}(H_i) \quad \text{for example} \quad z_i = \frac{1}{T_i} \sum_{t=1}^{T_i} h_{it}.$$

- **Static embeddings:** one vector per word type, regardless of context
- **Contextual embeddings:** one vector per occurrence; pool them into a sentence or document vector
- Images work in exactly the same way at the level we need:  $z_i = f_\theta(l_i)$

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# Outline

- 2 Text as Data
  - Representing Text
  - Topic Models

## Steps in Text Analysis

1. **Get the corpus.** Scraping, APIs, OCR. Usually the slowest part
2. **Preprocess.** Tokenize, lowercase, drop stopwords, stem or lemmatize, build the vocabulary
3. **Represent.** Turn documents into numbers: counts or embeddings
4. **Apply a method.** Dictionary, supervised, or unsupervised
5. **Validate.** Read the text. Always read the text
6. **Use it.** The estimate becomes a variable in an economic research design

Steps 1–2 are most of the work. Step 6 is where the paper is.

# Step 1: Representing Text as Data

- **Bag of words:** a document is a vector of counts over the vocabulary
  - Simple, transparent, and still competitive for many classification tasks
  - Weight by **tf-idf** to downweight words that appear everywhere
  - Throws away word order, syntax and synonymy
- **$n$ -grams:** count short phrases instead of single words. Recovers a little word order at the cost of a much larger vocabulary
- **Embeddings:** dense vectors  $z_i = f_\theta(r_i)$  from a pretrained encoder
  - Captures synonymy and context; no interpretable dimensions
- **Choose by task.** If you need to say *which words* drive a result, keep counts. If you need semantic similarity, use embeddings. Often: both, and check they agree

## Step 2: Statistical Methods

1. **Dictionary-based:**  $\hat{y}_i = f(x_i)$  for some known  $f$ 
  - Sentiment via a word list with scores; Loughran-McDonald for finance
  - Transparent and reproducible. Blind to context and easy to fool
2. **Supervised:** learn the mapping from words to an outcome  $Y$ , then predict  $\hat{Y}$ 
  - Lasso, forests, neural nets on counts or embeddings
  - Needs labels. Hand-code a few hundred documents, or use an LLM to label
3. **Unsupervised:** map words into latent variables  $\hat{\theta}$ , such as the topics in each document
  - Generative models: LDA and its descendants
  - No labels needed. Interpretation is entirely on you

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# Generative Language Models: Topic Models

- A topic model organizes a corpus into interpretable topics. LDA is the canonical version
- **Input:** bag-of-words documents, a number of topics  $K$ , tuning parameters
- **Idea:** words in document  $i$  are drawn from a document-specific distribution over the vocabulary,  $q_i$ , which is itself a mixture of topics:
  - $c_i \sim MN(\mathbf{q}_i, m_i)$  for document length  $m_i$
  - $\mathbf{q}_i = v_{i1}\beta_1 + \dots + v_{iK}\beta_K$
  - A topic  $\beta_k$  is a probability vector over words;  $\theta_i$  is document  $i$ 's distribution over topics
- **Output:** a distribution over words for each topic, and a distribution over topics for each document
- **Estimation:** variational inference, maximizing a bound on the likelihood (the ELBO)
- Modern variants embed the vocabulary first (embedded topic models), which handles large vocabularies far better



# The Fixed-Effects Problem in Topic Models

- Let  $\theta_i$  be true topic proportions and  $\mathbf{x}_i$  encode the environment (industry, valence, ...). A topic model fit on pooled data returns

$$\hat{\theta}_i = \theta_i + g(\mathbf{x}_i) + \nu_i$$

where  $g(\mathbf{x}_i)$  is systematic distortion from environment-specific language

- Now run the regression you actually care about, *with* environment fixed effects:

$$y_i = \alpha + \gamma \hat{\theta}_i + \delta^\top \mathbf{x}_i + \varepsilon_i$$

- Substituting:

$$y_i = \alpha + \gamma \theta_i + \underbrace{\gamma g(\mathbf{x}_i)}_{\text{bias}} + \gamma \nu_i + \delta^\top \mathbf{x}_i + \varepsilon_i$$

- Because  $g(\mathbf{x}_i)$  is correlated with  $\mathbf{x}_i$ , adding  $\delta^\top \mathbf{x}_i$  **does not absorb it**.  $\hat{\gamma}$  stays biased
- The lesson generalizes far beyond topic models:** if a measurement is contaminated by a covariate, controlling for that covariate ex post does not clean the measurement



# Mechanistic Fixed-Effects Topic Model (mFETM)

[Biderman, Blei, Cai, Ciccarone, Feder, Prat]

- Put both fixes together: fixed effects *in* the generative process, over a *concept* vocabulary
- **Representation**: 906,938 Glassdoor documents read by Gemma-2-9B, activations passed through a pretrained Gemma Scope SAE, pruned to  $V = 4,735$  labeled concepts
- **Environments**  $\mathbf{x}_i$ : 20 NAICS industries and 2 polarities, so  $E = 22$
- Generative process, per concept token  $j$  in document  $i$ :

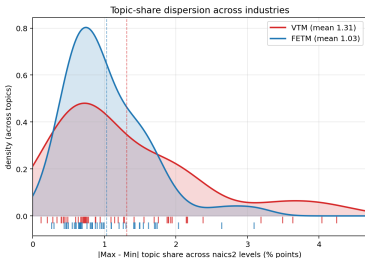
$$z_{ij} \sim \text{Cat}(\pi_i), \quad \pi_i \equiv \text{softmax}(\theta_i)$$

$$c_{ij} \sim \text{Cat}\left(\text{softmax}\left(\beta_{z_{ij}}^* + \gamma_{z_{ij}} \cdot \mathbf{x}_i\right)\right)$$

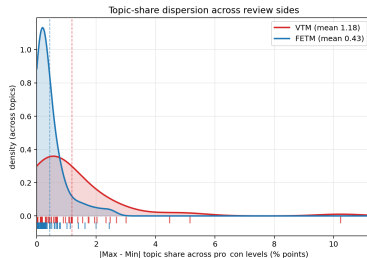
- $\beta_k^*$  is the **global backbone** of topic  $k$ ;  $\gamma_{e,k}$  is environment  $e$ 's **deviation** from it
- Identified by **weighted-centering**  $\gamma$  within each block, plus a shrinkage prior on the deviations. Setting  $\gamma \equiv 0$  recovers the pooled baseline (mVTM)

# Does It Work? Topic Portability

- If a topic meant to measure workplace culture also reveals whether the review was positive or negative, cross-firm comparisons inherit that
- Per topic, spread  $\Delta_k = \max_e \bar{\pi}_{ke} - \min_e \bar{\pi}_{ke}$  across environments. Lower is more portable



Source: Across industries



Source: Across pros vs. cons

- Industries: 1.31pp under mVTM  $\rightarrow$  1.03pp under mFETM ( $t = -1.72$ ,  $p = 0.090$ )
- Pros vs. cons: 1.18pp  $\rightarrow$  0.43pp, a reduction of nearly two thirds ( $t = -2.98$ ,  $p = 0.004$ )





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  - Why Images
  - Supervised Tasks
  - From CNNs to Foundation Models
  - Joint Text-Image Spaces

# Images and Social Science: a Long Way to Go

- Political and social scientists have long understood that visual data matters
- Before machine learning, analysis meant manual collection and annotation: expensive, small samples
- ML enables types of analysis that were simply unavailable before
- And the supply of data exploded. X (Twitter) alone produces roughly 3,000 images **per second**
- Still very little work using images, especially in economics
- **That gap is the opportunity.** The methods are no longer the binding constraint

# Text vs. Image

- Visual data differs from text in ways that change the workflow:
  - Text is high-dimensional (atomic unit: a word). An image is higher-dimensional still: an array of pixels, each with three RGB components
  - Text is **small**; images are **big**. Storage and compute become real constraints
  - Text is **language specific**; images are **universal**. Cross-country work is much easier
  - There is no natural “vocabulary” for images: no equivalent of counting words
- Historically, image analysis required a lot of work just to reach the starting point of text analysis
- **This is no longer true**, and that is the main update in this lecture

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## Supervised Tasks: Classification and Detection

- **Image classification**

- A  $Y = f(X)$  prediction problem, exactly as before
- Assigns a label  $y_i$  from a prespecified set
- The target need not be an object: scene and concept recognition work the same way

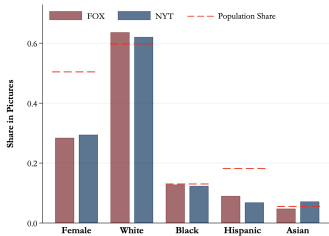
- **Object detection**

- Localize object instances *and* classify them
- Two steps: propose candidate regions, then classify each
- Useful when you care about counts or composition: how many people, whose faces, which flags

- Everything from the first block applies unchanged: train/validate/test, AUC, class imbalance

# Supervised Tasks: Face Analysis

- **Face detection:** locate faces in an image. A binary classification problem over sub-regions
- **Face recognition:** classify the identity of a face
- **Face analysis:** predict attributes such as expressed emotion, age, or perceived demographics



Notes: This figure shows the shares of different gender and ethnic groups in the pictures of Fox and NYT. The red lines indicate the share of each group in the U.S. population.

Attribute classifiers have documented differential error rates across groups. A mismeasured regressor correlated with the group is an inferential problem, not only a fairness one.

# Supervised Tasks: Visual Sentiment

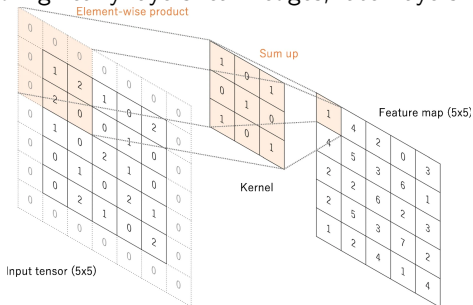
- Labeling emotion in images is subjective and hard to generalize
- Usually you must label yourself. Have many subjects do the task and average
- **Expressed emotion:** is the person in the image sad, happy, angry?
- **Evoked emotion:** does the image make the *viewer* feel sad, happy, angry?
- These are different constructs and they do not have to move together. Be explicit about which one your outcome variable is
- Modern shortcut: multimodal LLMs will rate both, cheaply. Validate against human labels on a subsample before you trust them at scale

# Outline

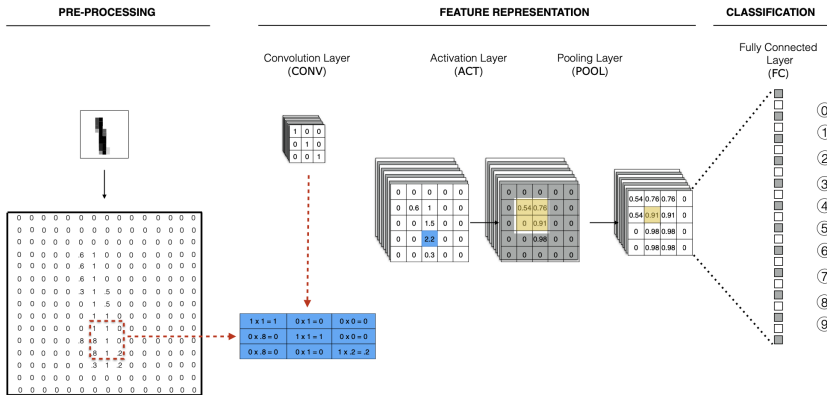
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# Convolutional Neural Networks

- A fully connected network on raw pixels has far too many parameters and ignores that nearby pixels are related
- A **convolution** slides a small learned filter across the image, producing a feature map
  - Same filter everywhere: massive parameter sharing, and translation invariance for free
- **Pooling** downsamples, making the representation robust to small shifts
- Stack convolution and pooling: early layers learn edges, later layers learn parts, then objects



# CNN Architecture: An Example



- Convolution and pooling blocks, then fully connected layers, then a softmax over classes
- **Remove the last layer and the penultimate activations are an image embedding.** This is the oldest trick in the applied literature and it still works

## How Do We Actually Train a CNN?

- In principle: same as any network. Cross-entropy loss, stochastic gradient descent, backpropagation
- In practice you almost never train from scratch. You do **transfer learning**:
  1. Take a model pretrained on a large corpus (ImageNet, or a web-scale image-text corpus)
  2. Freeze most of it
  3. Fit a small head on your few thousand labeled images
- This is why image work is now feasible with the sample sizes economists actually have
- **Data augmentation** (flips, crops, colour jitter) is regularization: it tells the model which transformations should not change the label



## Choosing an Approach in 2026

| Situation                                          | What to reach for                                                    |
|----------------------------------------------------|----------------------------------------------------------------------|
| You have a clear label and a few thousand examples | Fine-tune a pretrained model, or fit a head on frozen embeddings     |
| You have no labels and want structure              | Embed, then cluster                                                  |
| You want to construct a human could judge          | Multimodal LLM with a codebook, validated on a human-coded subsample |
| You want to compare images to text                 | A joint embedding space (next slide)                                 |
| You need experimental stimuli                      | Generative model, holding everything but one attribute fixed         |

Whatever you pick, the validation burden is the same.

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## Putting Text and Images in One Space: CLIP

- **CLIP** [Radford et al., 2021] embeds images and text into the **same vector space**
- Trained contrastively on 400M image-text pairs: maximize similarity for true pairs, minimize it for mismatched ones
- Result: a shared latent semantic space, up to a modality gap [Liang et al., 2022]
- Why this matters for research:
  - **Zero-shot classification**: score an image against text prompts, no training data needed
  - **Cross-modal comparison**: ask whether the picture and the words in the same message carry the same signal, on a common scale
  - **Concept scoring**: project images onto directions defined by words
- The last two are what make the following application possible at all

# Outline

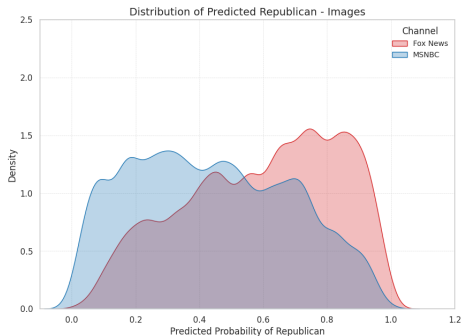
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## The Question

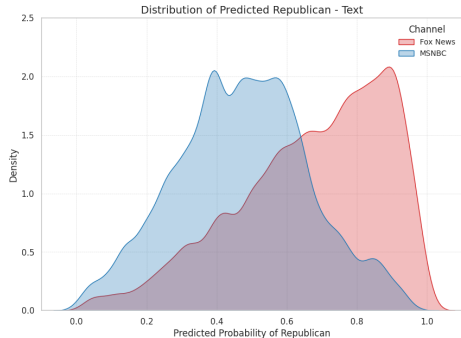
- Political news consumption is shifting to **video**, watched briefly and with low attention
- Almost everything we know about media slant is measured from **text** [Gentzkow and Shapiro (2010); Fowler et al. (2021)]
- If the image carries slant of its own, and carries it *faster*, then the move to video changes what political persuasion even is
- Two things are needed:
  1. An **ML framework** that quantifies image and text slant *separately*, at any exposure length
  2. A **survey experiment** that varies image and text content independently, to get from measurement to effects
- Today is about the first. But notice the shape: the measurement exists to make the experiment interpretable



# The Measure Transfers: Fox News vs. MSNBC



Source: Image model

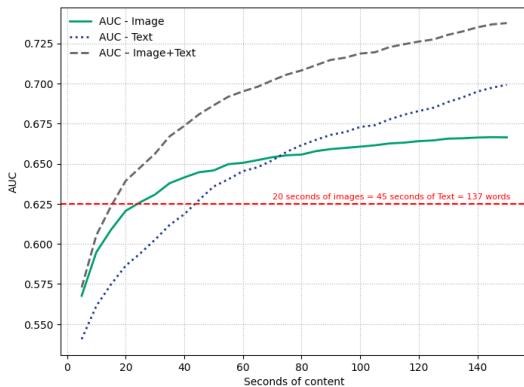


Source: Text model

- Trained on ads, applied to news video with no retraining, and it still separates the channels
- **This is the payoff of transfer learning for measurement:** labels existed in one corpus, the question lives in another

# Property 1: Speed of Information Transfer

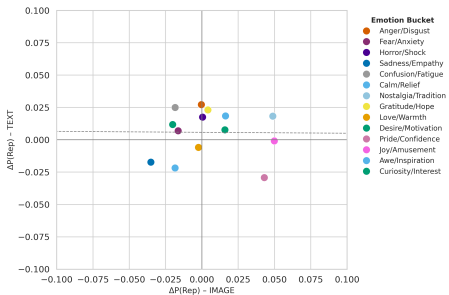
Sample  $K$  random 5-second chunks per video, compute AUC, and compare across exposure lengths.



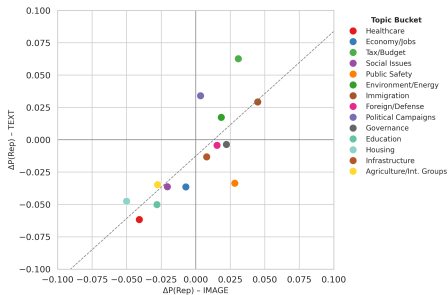
- The image signal saturates **much faster** than the text signal
- A viewer who watches for a few seconds has received the visual slant almost in full, and very little of the textual slant



# Images Embed Emotions; Text Embeds Substantive Topics



Source: Emotion concepts



Source: Topic concepts

- **Emotions:** mean absolute CTE 0.021 for image vs. 0.016 for text, and the two are *uncorrelated* ( $r = -0.01$ )
- **Topics:** 0.025 for image vs. 0.033 for text, and here they *do* agree ( $r = 0.76$ )
- So the modalities converge on substance and diverge on affect. Images are not fast text



1. **A measurement problem:** partisanship is not observed, only party labels on ads
2. **A representation:** CLIP embeddings, chosen because they let text and images be compared on one scale
3. **A supervised model**, evaluated honestly out of sample
4. **Transfer:** apply the trained measure to a corpus where no labels exist (news video)
5. **An interpretability layer**, so the measure is not a black box and yields testable properties
6. **Economics:** an experiment those properties made it possible to design and interpret

Steps 1–5 are this course. Step 6 is your paper. Do not stop at step 3.

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## Practical Guidance

- **Start small.** Build the whole pipeline on 500 documents or images before scaling. Every pipeline is wrong the first time
- **Version everything.** Model name, model version, prompt, preprocessing. Embedding models are updated and deprecated, and your results move with them
- **Cache aggressively.** Embeddings and API labels are expensive to recompute and trivial to store
- **Budget before you run.** Price the full job on a pilot sample first
- **Hand-code a validation sample.** A few hundred documents you have read yourself is the single highest-return thing you can do
- **Report measurement error.** Your regressor is an estimate. Say so, and show the result survives a noisier or cleaner version of it





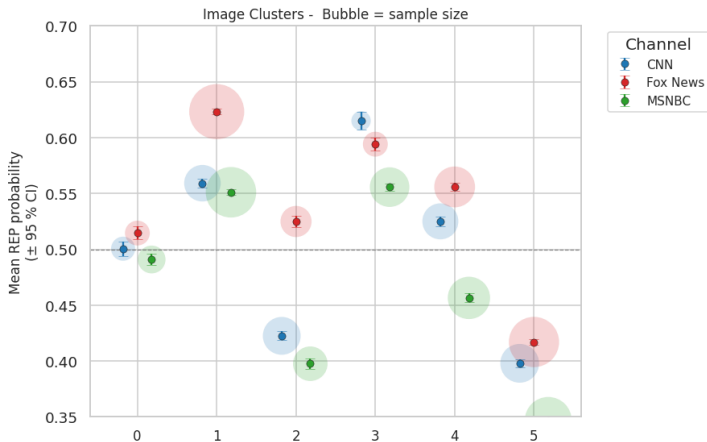
## Appendix: Generative Models

- **GANs**: a generator creates synthetic images while a discriminator tries to tell real from fake. At equilibrium the discriminator cannot distinguish them
- Largely superseded by **diffusion models**, which learn to reverse a gradual noising process. More stable to train and far better quality
- Research use is mainly **experimental stimuli**: generate images identical except along one attribute, then randomize
- Validity caveat: generated counterfactuals often change more than the attribute you intended. Check before you interpret the treatment effect



(a) Side-by-side mugshot detention morphs with detention probabilities of 0.41 and 0.13 respectively

## Appendix: Image Clusters in News Video



- Unsupervised structure in news frames, before any partisanship label is applied